

COLDPOD · DESIGN DECISIONS REGISTER

ColdPod design decisions register

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DOCUMENT CONTROL

Document ID	CPD-DEC-001	Version	0.4
Status	Draft	Date	2026-10-02
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/coldpod		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-02	Amish Chadha	Value engineering restated after the approved follow-ups were carried into the model and BOM: USD 313, USD 3 over the target (battery shipping switch added)
0.3	2026-10-02	Amish Chadha	Amish approved the recommendations for open decisions 1 to 7 (2026-10-02); moved to decisions made (CPD-DDR-003 accepted)
0.2	2026-10-01	Amish Chadha	Budget treated as a value-engineering target
0.1	2026-10-01	Amish Chadha	Register opened with the open decisions from the review note, the decision records and the build plan work

ColdPod design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

None. All open decisions were decided on 2026-10-02.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The VIP maker can supply the seven panels to plus 0, minus 1 mm, with a quote and aged conductivity and edge data	The panels fill the shell exactly; joints and edges are about 45 % of the heat leak	CPD-DDR-003; CPD-CAL-001
2	The gasket still seals with the lid tray landed on the can flange, and the tray seat gives about 0.5 K/W	Lid seal and lid-pack refreeze both rest on it	CPD-DDR-002; CPD-CAL-001
3	The PCM maker can supply the eight pouch shapes, and the usable latent heat between 2 and 8 °C	Pouch fit round the risers and feet; hold times	CPD-DDR-003; CPD-CAL-001
4	The bought handle's pivot eyes take M5 screws and clear the 4 mm pads	Handle pivot pads	CPD-DDR-003
5	The off-state conductance and real COP of the chosen module with the chosen heat sink and fan	Hold power and the thermosiphon diode ratio	CPD-PRC-001
6	The Peltier driver can run the module from a 10 V input	R7 at the low end of the vehicle range	CPD-DDR-002
7	Whether a motorbike carrier stays within about 19° of tilt with the cooling head down; if not, the mechanical disconnect fallback	The loop thermosiphon works only below that tilt	CPD-DDR-001, D2; CPD-CAL-001, H5

■ Value engineering

Value-engineering target: USD 310 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 313 (USD 3 over the target).

- **Main cost drivers:** the vacuum-insulated panel set (about USD 60, still an estimate until quoted), the thermosiphon loop (about USD 28), the buck-boost Peltier driver (USD 22 to 26), the lid cold plate tray (USD 13) and the liner cut-out (USD 10 to 13).
- **Savings worth trying:** a VIP quote before any purchase; thinner printed parts, which already saved about USD 4 of filament; and re-pricing the driver and thermosiphon parts at purchase; the USD 5 shipping switch could wait until a trip needs it, which would bring the cost back to USD 308.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D11: 5 °C organic PCM, thermosiphon with a mechanical disconnect as fallback, evaporator on the outer face of the PCM, 76.8 Wh LiFePO4 pack, R6 redefined to 16 h, R12 relaxed to 5.5 kg, VIPs with a foam variant documented, local alarms and Bluetooth only, alarm defaults, budget \$300, outreach vaccinators first	Amish, going with the recommendation	CPD-DDR-001
2026-09-25	TRL 3 recommendations: lid cold plate, 3 °C liner cut-out, thinner printed parts, evaporator can and loop thermosiphon, TEC1-12703 class with smooth DC, buck-boost driver, budget kept at \$300	Amish: "i accept all your recommendations, go with them across all repos."	CPD-DDR-002
2026-09-26	Budget top-up to \$310 (R15 met)	Amish: "I am ok with the budget top ups"	CPD-DDR-002, N3
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." The changes themselves were accepted on 2026-10-02 (below)	CPD-DDR-003
2026-10-02	Design for construction accepted: the changes P1 to P12 (pads and towers on the shell, rim slots, foam strip, loop in the can corner, block frame, liner feet and collar, folded lid tray, fixings, drip tray, cable duct) and their knock-on changes, as made	Amish: "i approve your recommendations for all 555 open decisions."	CPD-DDR-003, A1
2026-10-02	Mass (R12): option (a). R12 relaxed to 5.75 kg for the prototype, which is weighed at TRL 4; the base bumper is left off so the 0.05 kg margin is not spent before weighing	Amish: "i approve your recommendations for all 555 open decisions."	CPD-DDR-003, A2; CPD-DDR-002, N2
2026-10-02	Refreeze (R8): option (a). R8 restated as 9 h or less for all the PCM at 25 °C, still an overnight refreeze	Amish: "i approve your recommendations for all 555 open decisions."	CPD-DDR-002, N1; CPD-CAL-001 v0.4, G4
2026-10-02	Battery removal (R14): option (a) for the prototype; the battery bay on four screws counts as removable. Before any field or air travel use, add the shipping switch and check the carrier's lithium battery rules. The switch is now in the model and the BOM (line 18, USD 5), which took the cost from USD 308 to USD 313	Amish: "i approve your recommendations for all 555 open decisions."	CPD-DDR-003, A3
2026-10-02	Loop working fluid: R-134a (non-flammable, safety class A1) for the prototype, with the charge set by the refrigeration technician from the loop volume;	Amish: "i approve your recommendations for all 555 open decisions."	CPD-DDR-003, A4

Date	Decision	Decided by	Record
	a lower-impact fluid such as R-1234yf (A2L, mildly flammable) is revisited only for a product version and under local rules		
2026-10-02	Co-design partner: an immunization program, consistent with outreach vaccinators first (CPD-DDR-001, D11). First candidate to approach: PATH, or a national immunization program's outreach team reached through it	Amish: "i approve your recommendations for all 555 open decisions."	CPD-DDR-001, O1
2026-10-02	Appearance model: round bail in the model; heat sink kept solid (no window); second status light adopted; labels and wordmark under BOM line 16; base bumper left out until the prototype is weighed	Amish: "i approve your recommendations for all 555 open decisions."	docs/ REVIEW.md , 2026-09-26